

TOPIC

10

Legacy matrices (removed from current syllabus)

Legacy practice only | Not part of the current syllabus

2

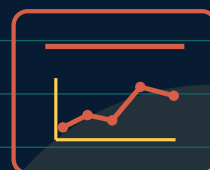
QUESTIONS

2

EXAM PAGES

1

SECTIONS



TOPIC CONTENTS

1/1

Quick navigation and progress overview

10.1

Matrices (legacy content)

2 questions | 2 pages

SECTION 10.1

Matrices (legacy content)

Sorted by session, paper variant and original question number

2

QUESTIONS

2

EXAM PAGES

P4

CALCULATOR



10.1 Matrices (legacy content)

#	SOURCE QUESTION	PAGES	DONE	MARK	REVIEW
01	2015 May June Paper 43 Question 9	1	☐	—	☐
02	2016 May June Paper 43 Question 8	1	☐	—	☐

9 $\mathbf{P} = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ $\mathbf{Q} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ $\mathbf{R} = \begin{pmatrix} 0 & u \\ 1 & v \end{pmatrix}$ $\mathbf{S} = \begin{pmatrix} w & 3 \\ 8 & 2 \end{pmatrix}$

(a) Work out $\mathbf{PQ}$.

Answer(a) $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

(b) Find $\mathbf{Q}^{-1}$.

Answer(b) $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

(c) $\mathbf{PR} = \mathbf{RP}$

Find the value of u and the value of v .

Answer(c) $u = \dots\dots\dots$

$v = \dots\dots\dots$ [3]

(d) The determinant of $\mathbf{S}$ is 0.

Find the value of w .

Answer(d) $w = \dots\dots\dots$ [2]

8

$$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ -1 & 5 \\ 3 & -4 \end{pmatrix}$$

$$\mathbf{B} = \begin{pmatrix} 1 & 3 \\ -1 & 5 \end{pmatrix}$$

$$\mathbf{C} = \begin{pmatrix} 7 \\ -4 \end{pmatrix}$$

$$\mathbf{D} = \begin{pmatrix} 2 & 5 \end{pmatrix}$$

- (a) Work out each of the following if the answer is possible.
If a calculation is not possible, write “not possible” in the answer space.

(i) $\mathbf{BA}$

[1]

(ii) $2\mathbf{A}$

[1]

(iii) $\mathbf{CD}$

[2]

(iv) $\mathbf{DC}$

[2]

(v) $\mathbf{B}^2$

[2]

- (b) Find $\mathbf{B}^{-1}$, the inverse of $\mathbf{B}$.

$\begin{pmatrix} & \\ & \end{pmatrix}$ [2]